

Introduction of Machine Learning

Disclaimer:

This PPT is modified based on:

- Dr. Hung-yi Lee
- http://speech.ee.ntu.edu.tw/~tlkagk/courses_ML17.html
- Quang Nhat NGUYEN

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What is Machine Learning?

- Machine Learning (ML) is a traditional and very active field of Artificial Intelligence (AI)
- Some examples of definition of ML
 - A process by that a system improves its performance [Simon, 1983]
 - A process by that a computer program improves its performance in a task through experience [Mitchell, 1997]
 - A programming of computers to improve a performance criterion based on past sample data or experience [Alpaydin, 2004]
- Representation of a ML problem [Mitchell, 1997]

ML = Improvement of a task's efficiency through experience

 - A task T
 - For the evaluation criteria of performance P
 - By using some experience E



What is Machine Learning

The subfield of computer science that *“gives computers the ability to learn without being explicitly programmed”*. (Arthur Samuel, 1959)

Using previous data for answering future questions

Training

Prediction

Historic may contain answers or may not contain answers

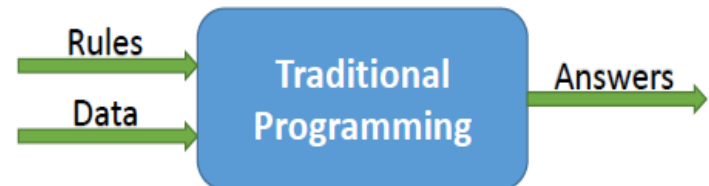
Labeled

Unlabeled

Machine Learning vs. Traditional Programming

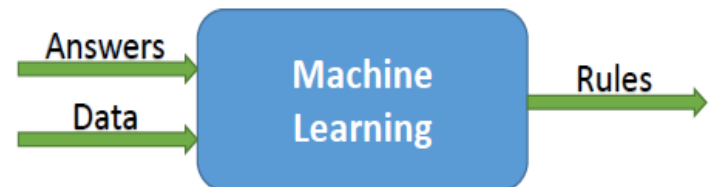
Traditional Programming:

- Business requirements and data are analyzed.
- A set of hard-coded rules are programmed and tested.
- Program process new data based on the coded rules.



Machine Learning:

- Data and their labels (answers) are fed into a model.
- Model "learns" useful features and frequent patterns to predict answers.
- Trained model is used to "predict" answers for new data.



Example - Activity Recognition



```
if(speed<4){  
    status=WALKING;  
}
```

```
if(speed<4){  
    status=WALKING;  
} else {  
    status=RUNNING;  
}
```

```
if(speed<4){  
    status=WALKING;  
} else if(speed<12){  
    status=RUNNING;  
} else {  
    status=BIKING;  
}
```

// Oh crap

Example - Activity Recognition - ML - Labels



```
0101001010100101010
1001010101001011101
0100101010010101001
0101001010100101010
```

Label = WALKING



```
1010100101001010101
0101010010010010001
0010011111010101111
1010100100111101011
```

Label = RUNNING



```
1001010011111010101
1101010111010101110
1010101111010101011
1111110001111010101
```

Label = BIKING



```
1111111111010011101
0011111010111110101
0101110101010101110
1010101010100111110
```

Label = GOLFING
(Sort of)

Example



HOUSING PRICES

Estimate housing prices based on 3 **features** (properties):

Area of the House, Number of Bedrooms, Number of Bathrooms

- **Traditional Approach:**

Price = 1.2 x Area + 0.7 x # Bedrooms + 0.3 x # Bathrooms

Pricing formula is known beforehand and is explicitly hard-coded. The formula can be deducted by manual analysis or SME domain experience.

- **Machine Learning:**

Price = A x Area + B x # Bedrooms + C x # Bathrooms

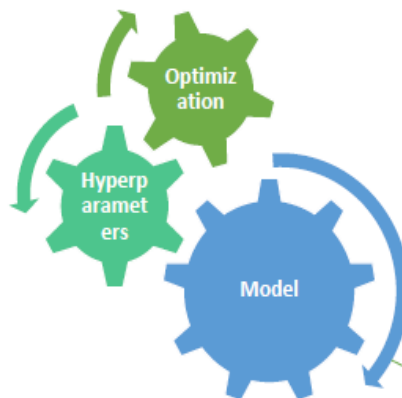
Pricing Formula is unknown at the beginning and would need the model to be trained to “Learn” the formula attributes A,B and C.

Example

Housing Prices

Area	#Bedrooms	#Bathrooms	Price
130	3	1	1,200
160	3	2	1,500
90	2	1	900
...			

Input Dataset:
Contains prepared
historic data of
actual house sales.



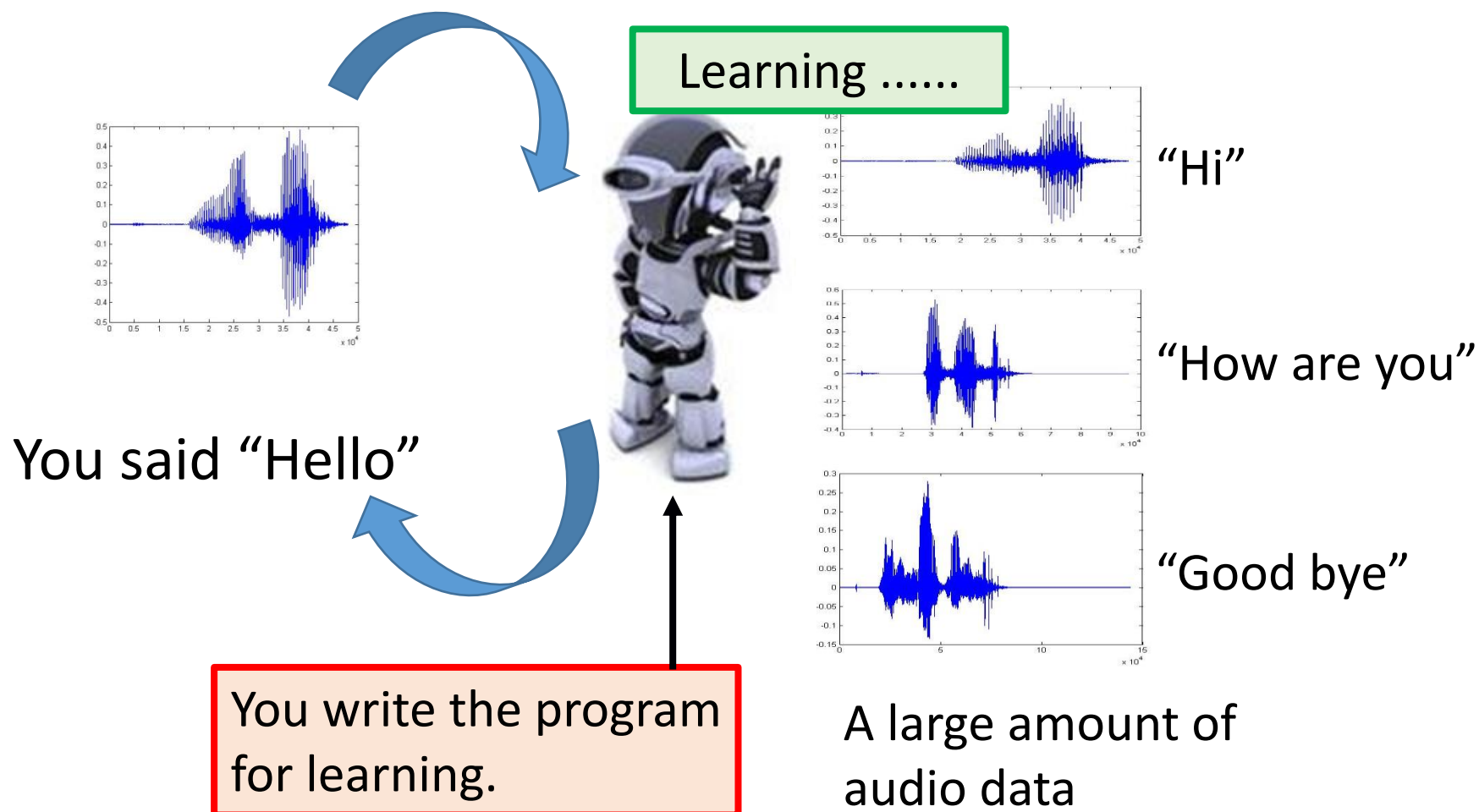
Model:
Model Learns appropriate
“Parameters” to “Fit” the
input data.

$$\begin{aligned} \text{Price} &= A \times \text{Area} + B \times \# \text{ Bedrooms} \\ &+ C \times \# \text{ Bathrooms} \\ A &= 1.247 \\ B &= 0.682 \\ C &= 0.319 \end{aligned}$$

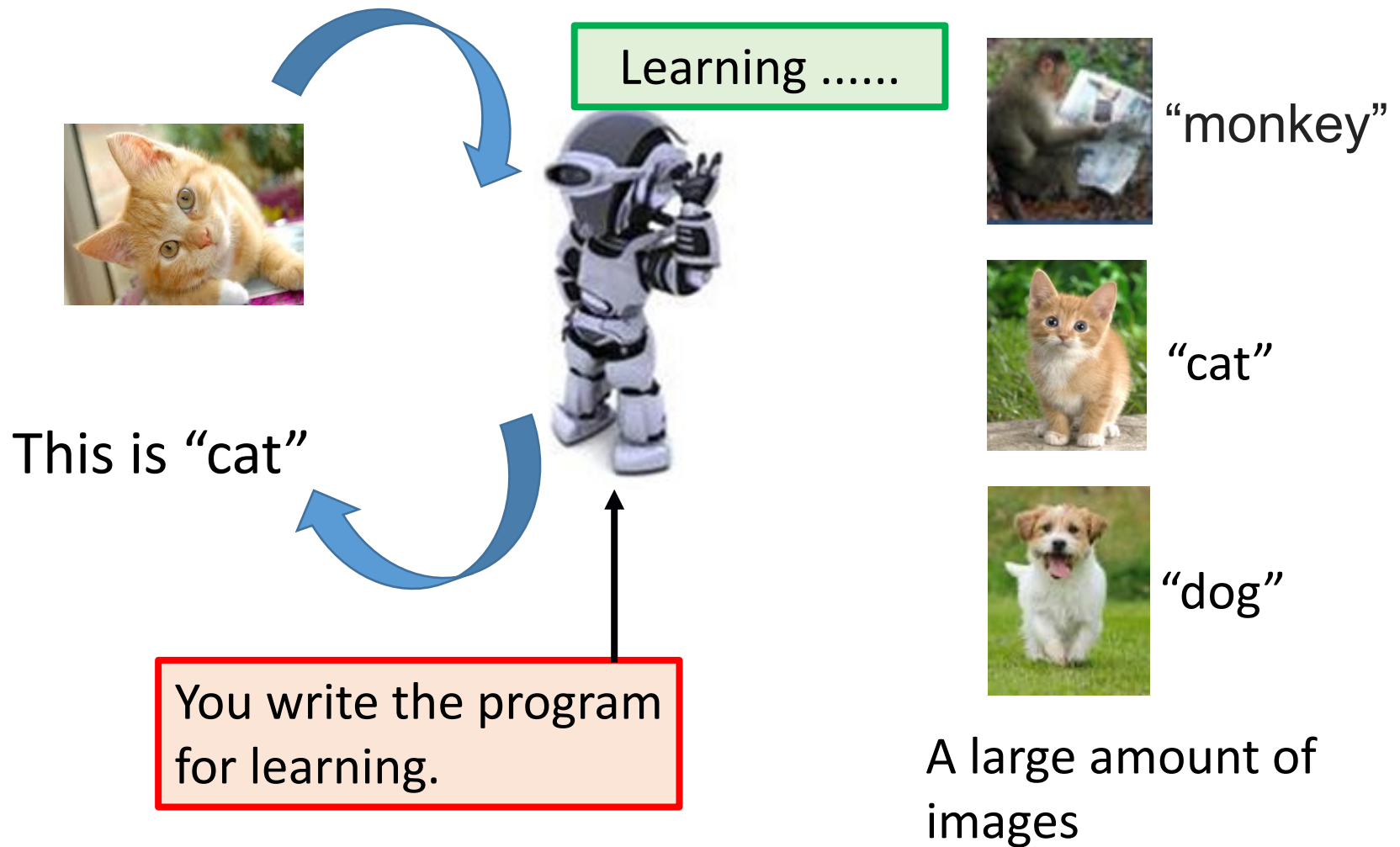
Output:
Parameters that completes the
formula and can be generalized to
predict unsold houses.



What is Machine Learning (speech recognition)?



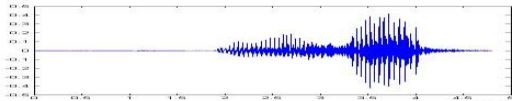
What is Machine Learning?



Machine Learning

≈ Looking for a Function

- Speech Recognition

$$f(\text{ ) = \text{“How are you”}$$

- Image Recognition

$$f(\text{ ) = \text{“Cat”}$$

- Playing Go

$$f(\text{ ) = \text{“5-5” (next move)}$$

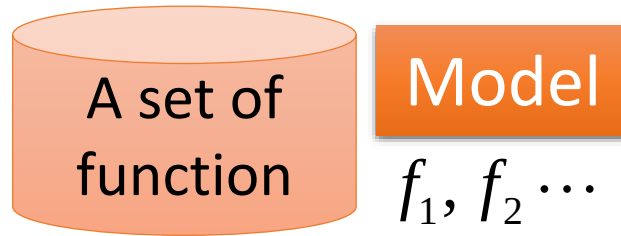
- Dialogue System

$$f(\text{“Hi” (what the user said) }) = \text{“Hello” (system response)}$$

Framework

Image Recognition:

$$f\left(\text{img}\right) = \text{"cat"}$$



$$f_1\left(\text{img}\right) = \text{"cat"}$$

$$f_2\left(\text{img}\right) = \text{"monkey"}$$

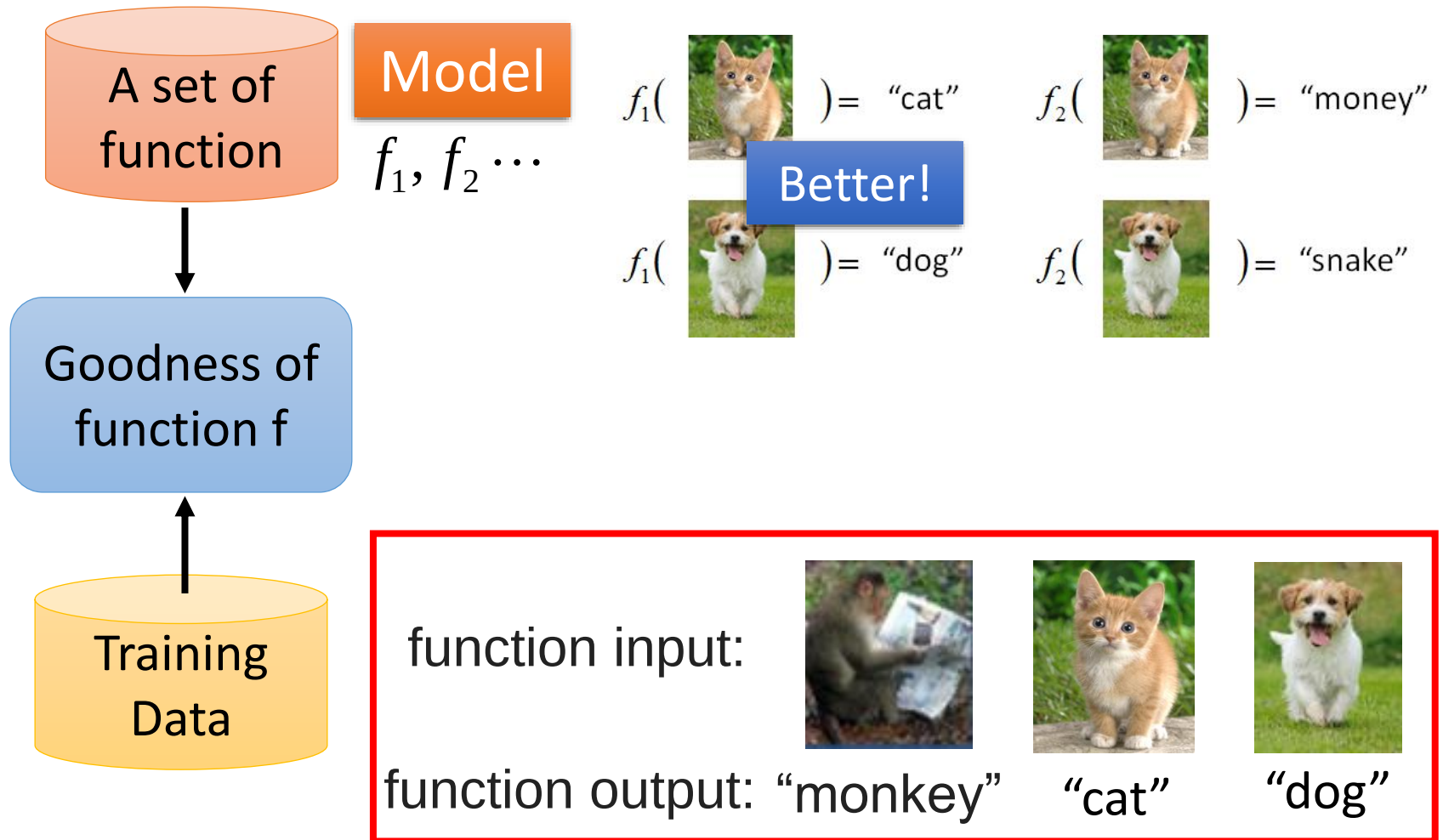
$$f_1\left(\text{img}\right) = \text{"dog"}$$

$$f_2\left(\text{img}\right) = \text{"snake"}$$

Framework

Image Recognition:

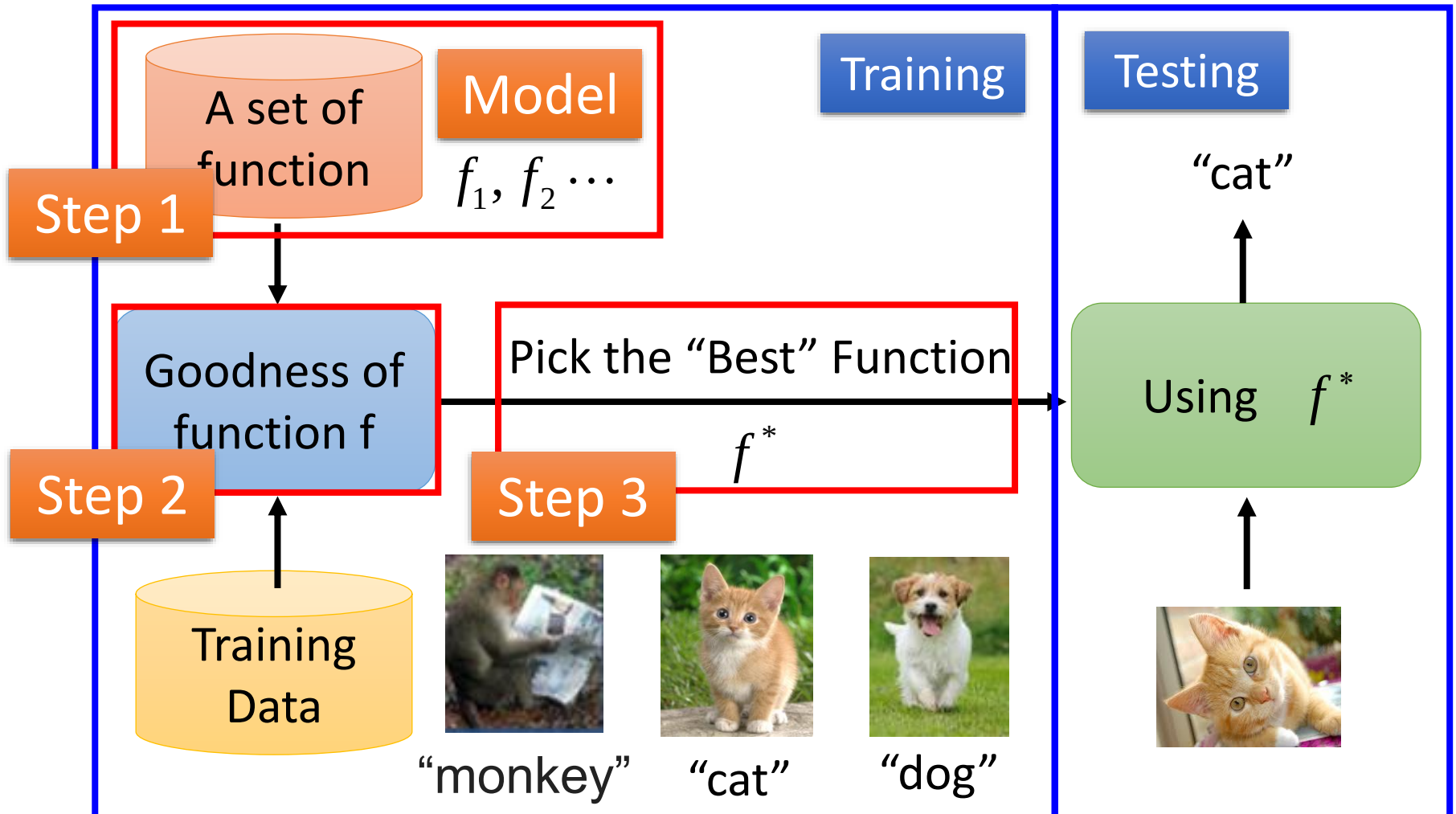
$$f\left(\text{img_cat}\right) = \text{"cat"}$$



Framework

Image Recognition:

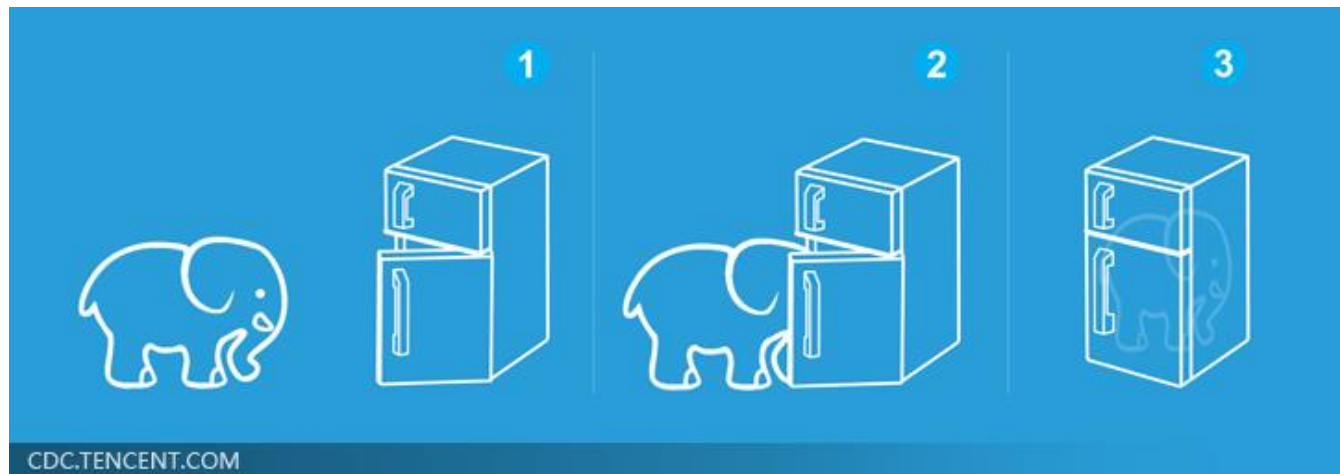
$$f\left(\text{Image of a cat}\right) = \text{"cat"}$$



Machine Learning is so simple



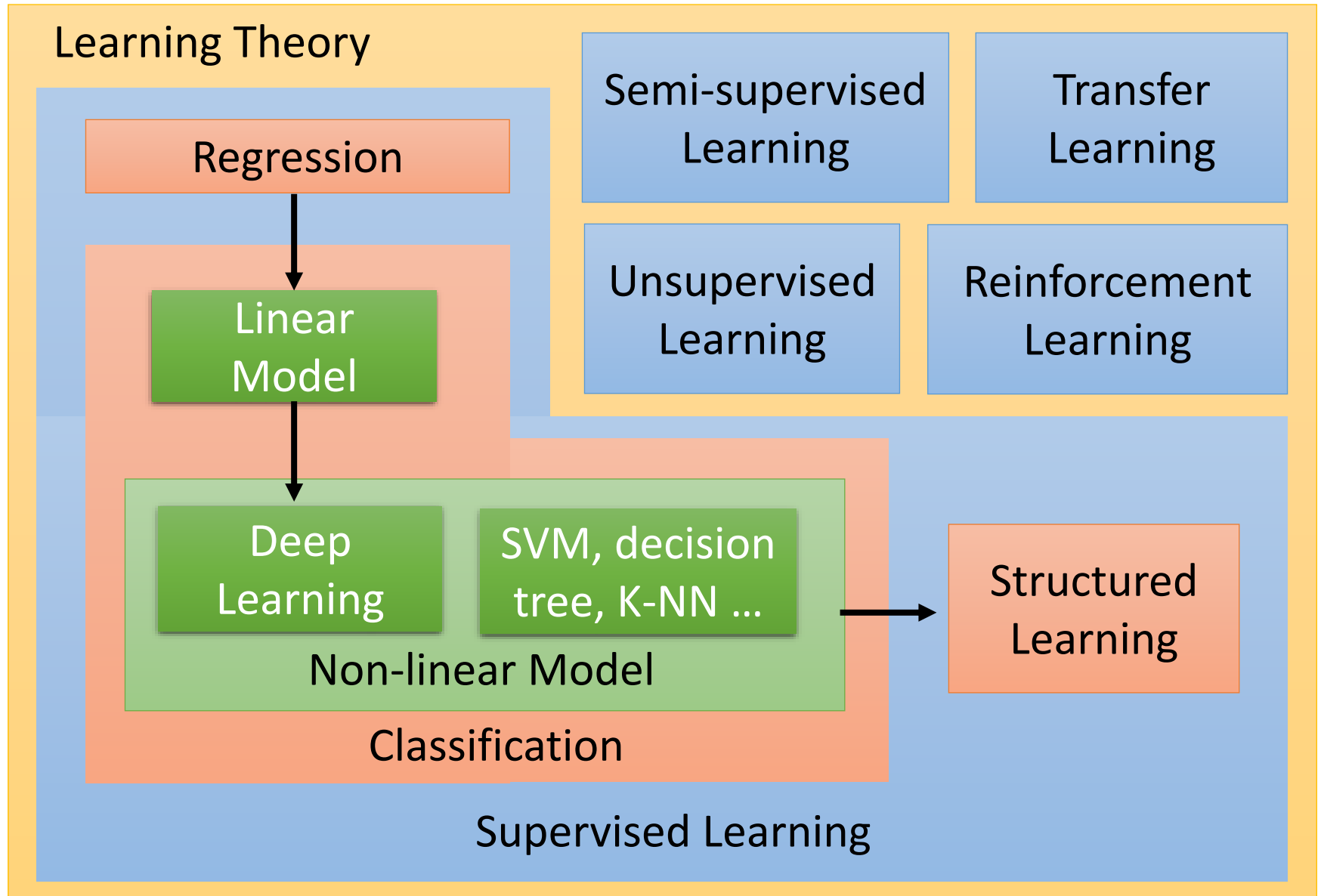
Similar to put an elephant into a refrigerator



Types of Machine Learning

Learning Map

scenario task method





Supervised Learning



Supervised

- Learn through examples collected from historic data.
- Examples contain the desired output (labels) that will be predicted for future data.

Unsupervised

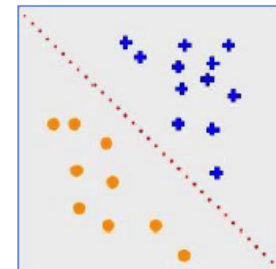
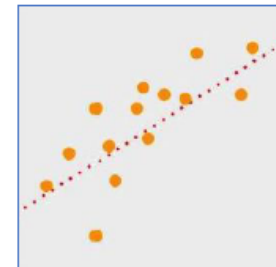
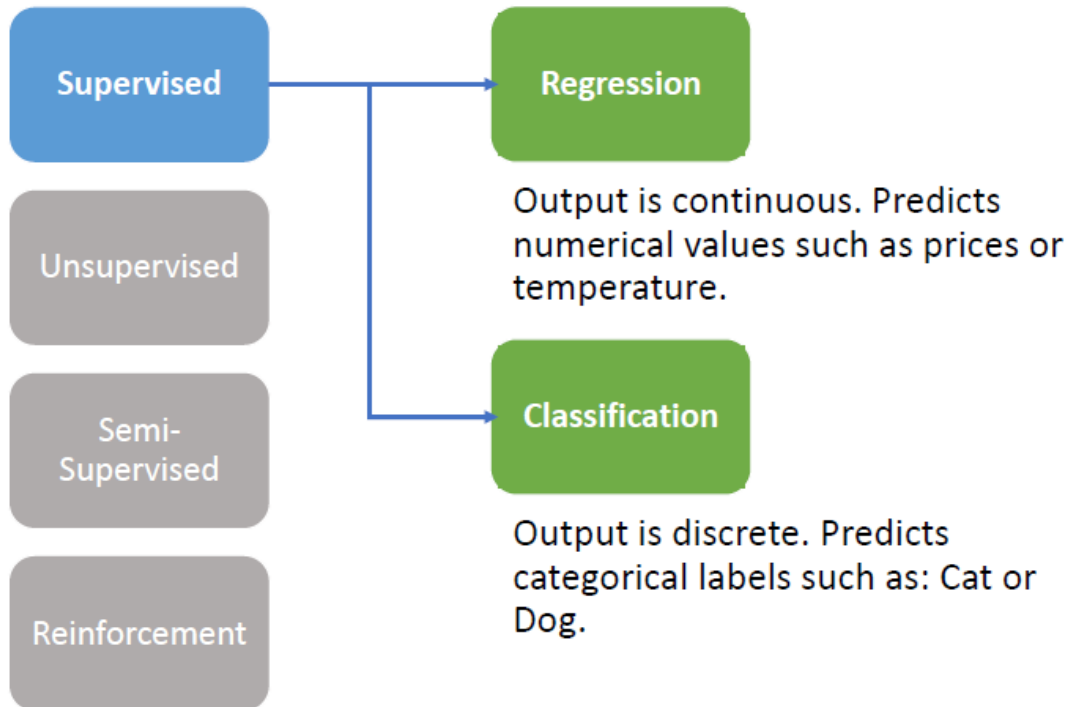
Semi-Supervised



Reinforcement

- Is this a cat or a dog?
- Is this email a spam or not?
- What is the market value of a house given its area and number of bedrooms?

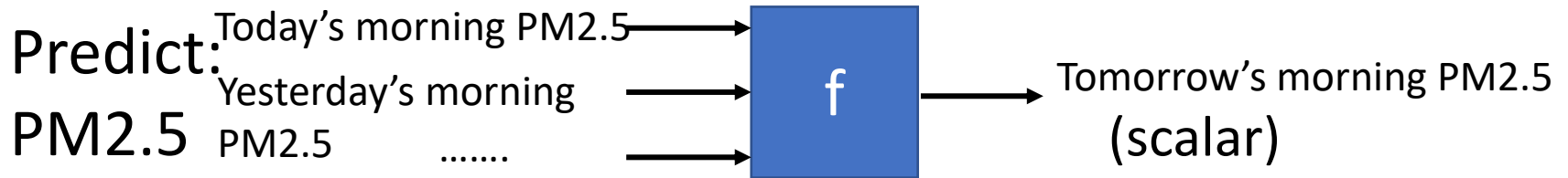
Supervised Learning



Learning Map

Regression

The output of the target function f is “scalar”.



Training Data:

Input:

9/01 morning PM2.5 = 63 9/02 morning PM2.5 = 65

Output:

9/03 morning PM2.5 = 100

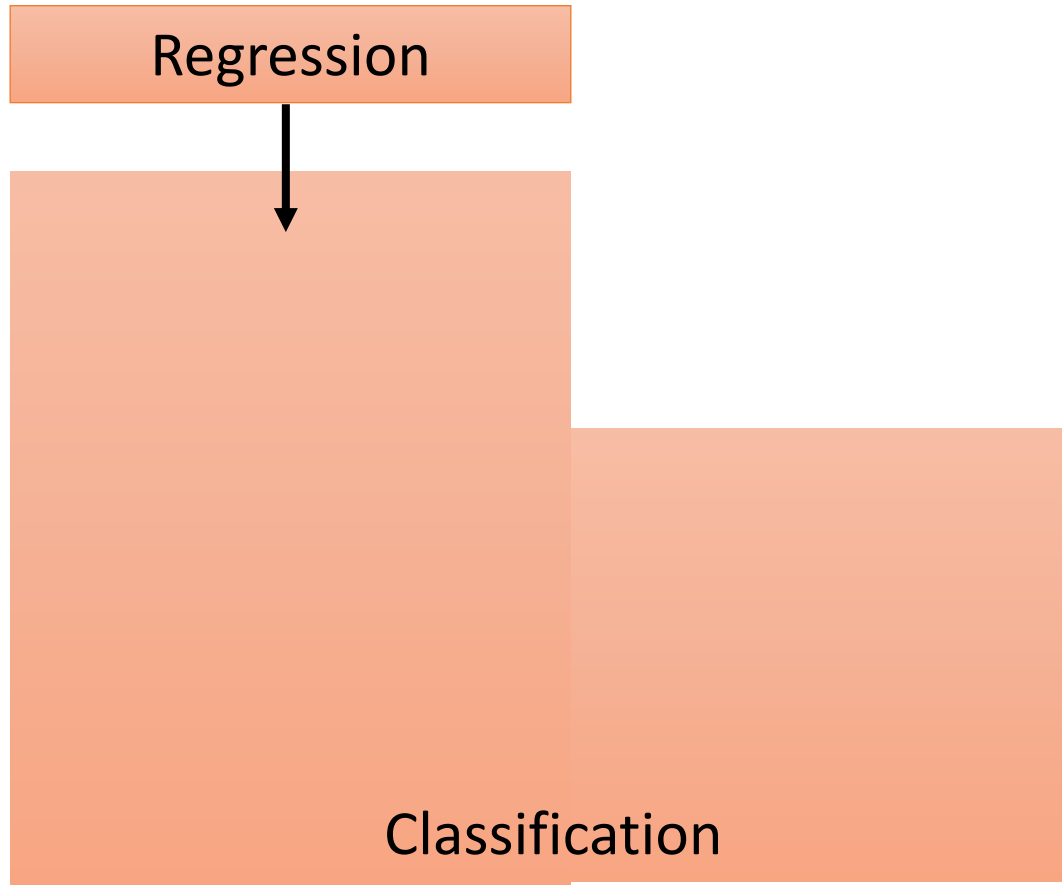
Input:

9/12 morning PM2.5 = 30 9/13 morning PM2.5 = 25

Output:

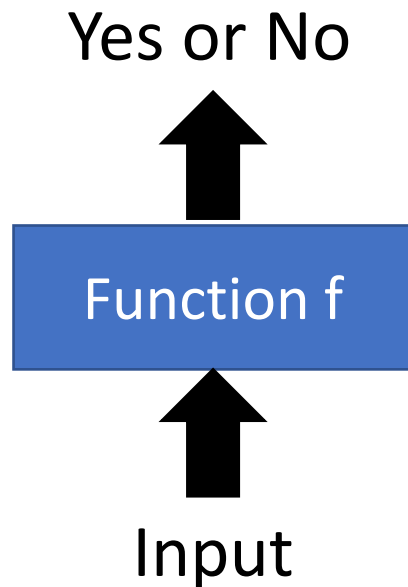
9/14 morning PM2.5 = 20

Learning Map

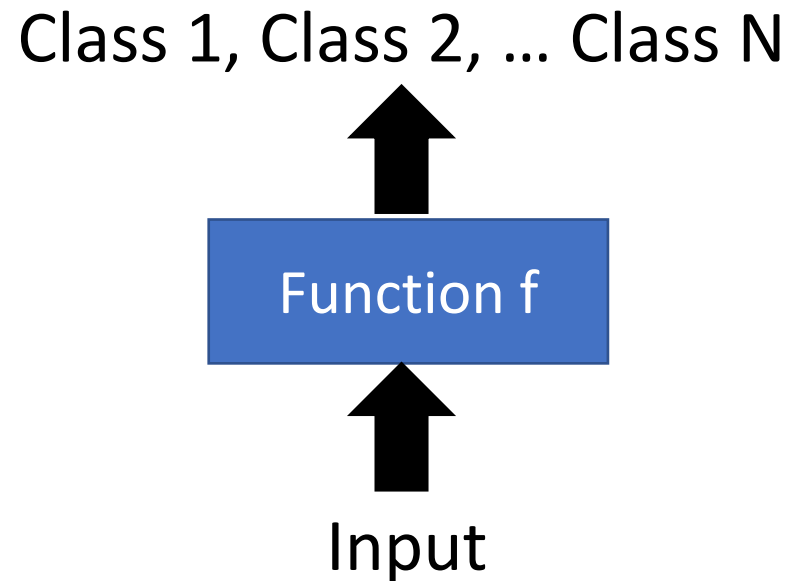


Classification

- Binary Classification

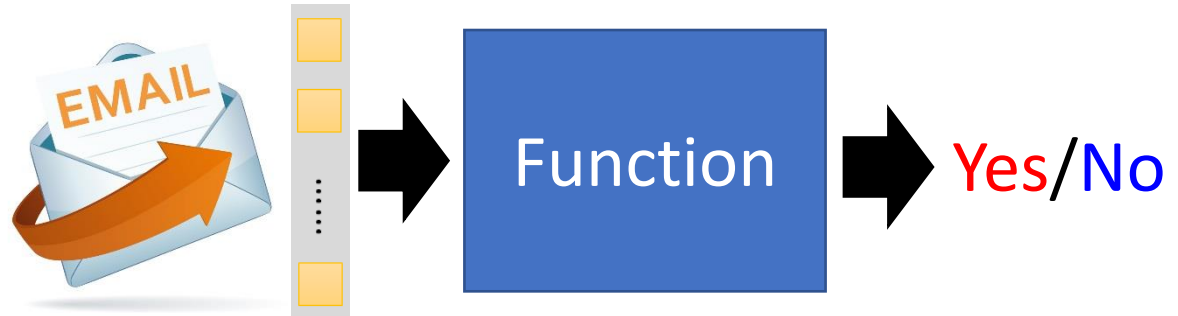


- Multi-class Classification



Binary Classification

Spam
filtering

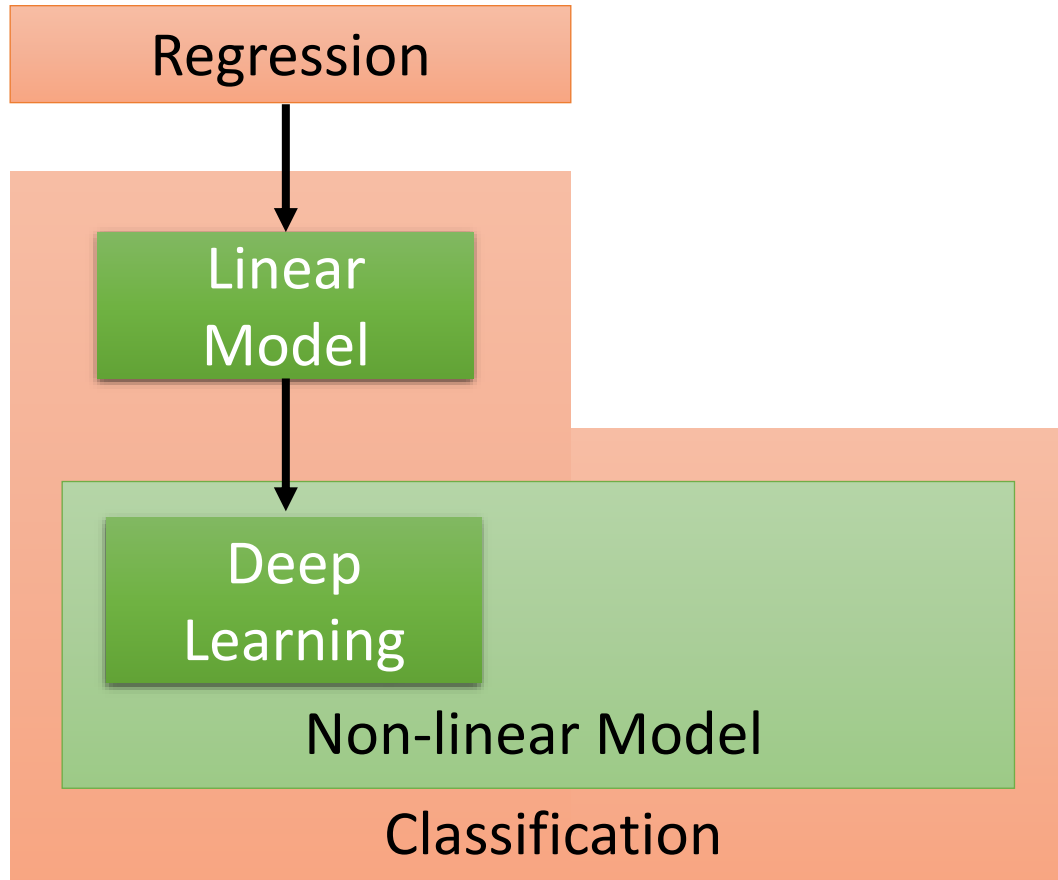


Multi-class Classification

Document Classification

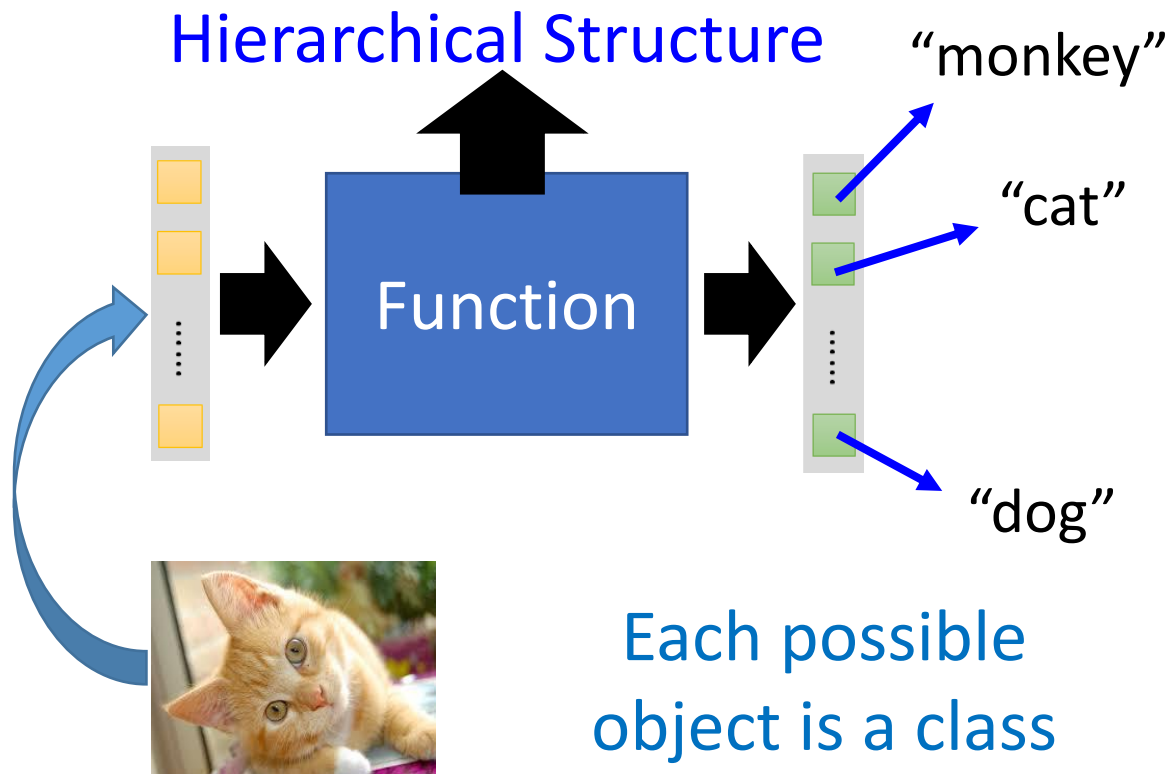


Learning Map



Classification - Deep Learning

- Image Recognition



Training Data



“monkey”



“cat”



“dog”

Classification - Deep Learning

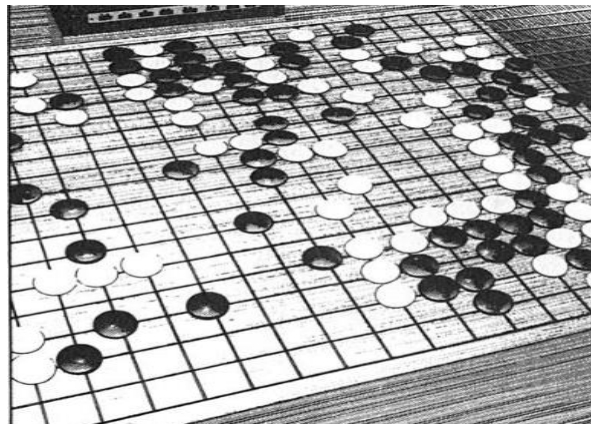
- Playing GO



Function

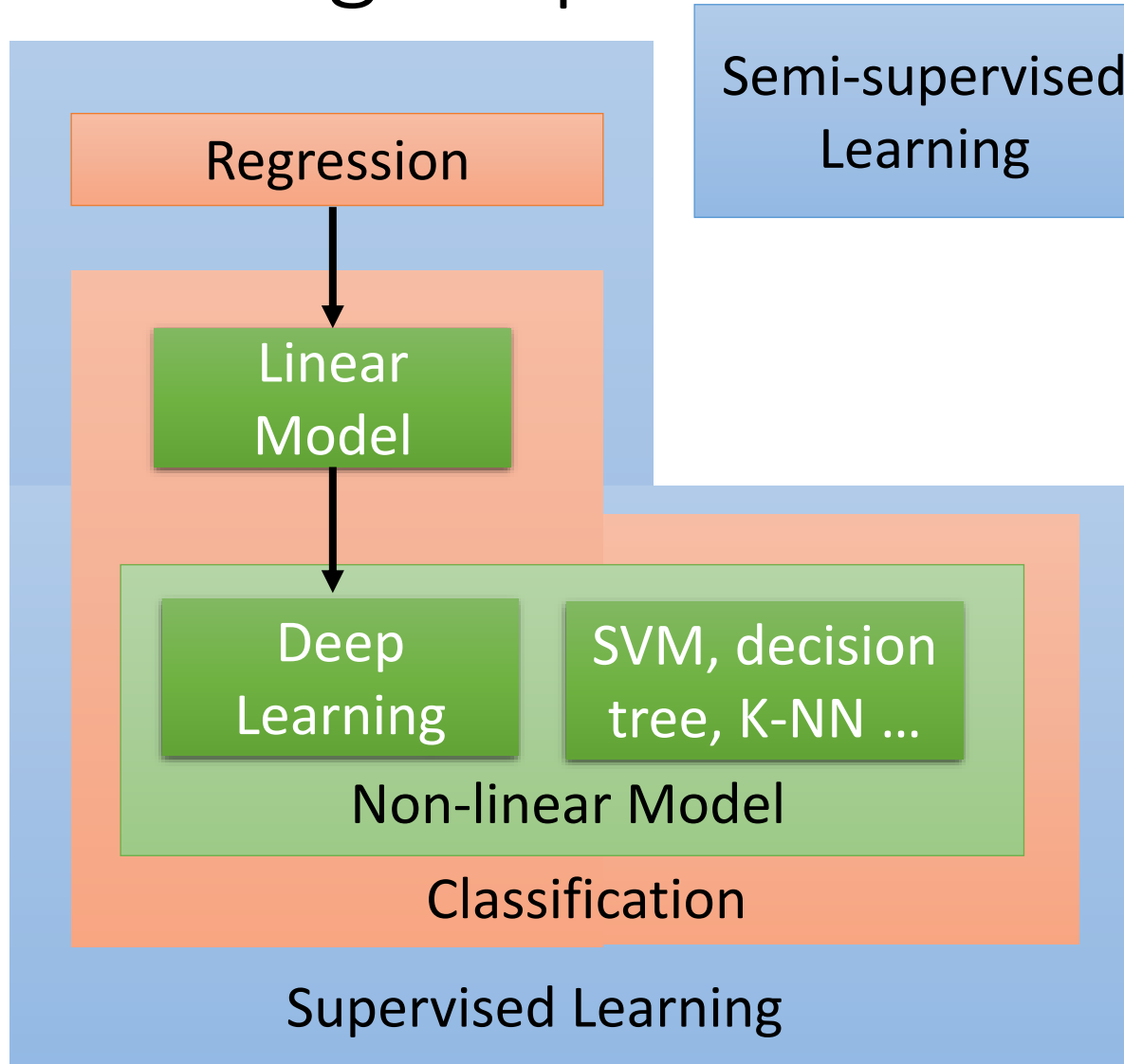
Next move
Each position
is a class
(19 x 19 classes)

Training Data



Learning Map

Hard to collect a large
amount of labelled data



Training Data:

Input/output
pair of target
function

Function
output = label

Hard to collect a large
amount of labelled data



Semi-Supervised Learning



Supervised

Unsupervised

Semi-
Supervised



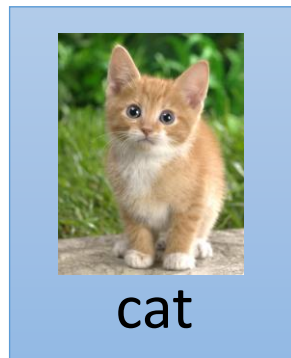
Reinforcement

- Historic data has a small amount of labeled data, and a large amount of unlabeled data.
- The cost of manually labeling all data is prohibitive.
- The problem is initially treated as Unsupervised to group data to different structure.
- After that, available labels are used to label entire clusters.

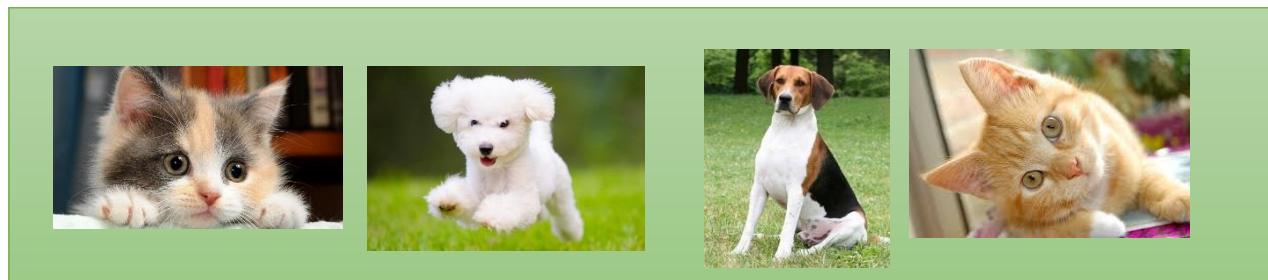
Semi-supervised Learning

For example, recognizing cats and dogs

Labelled
data

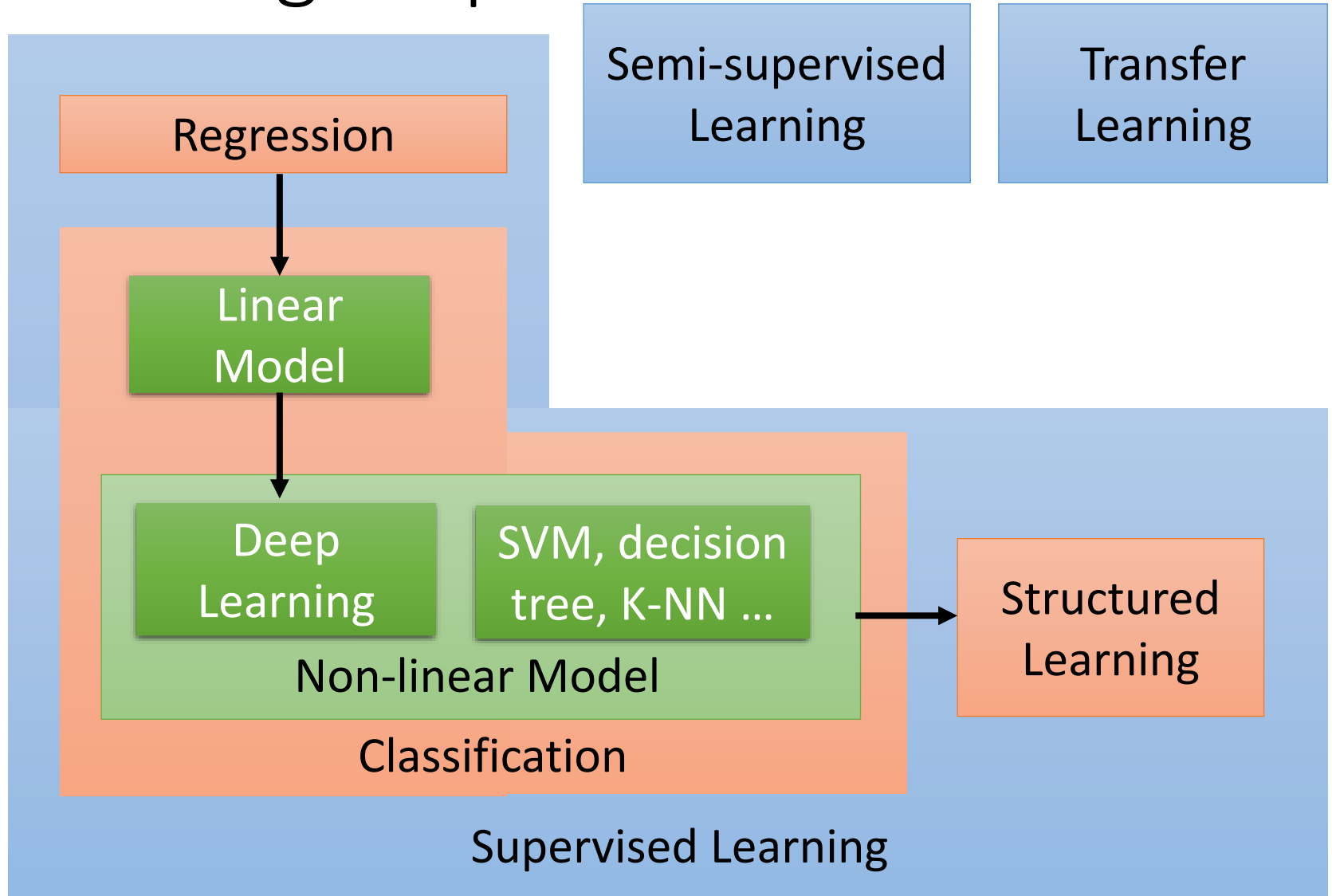


Unlabeled
data



(Images of cats and dogs)

Learning Map



Transfer Learning

For example, recognizing cats and dogs

Labelled
data



cat



dog



elephant

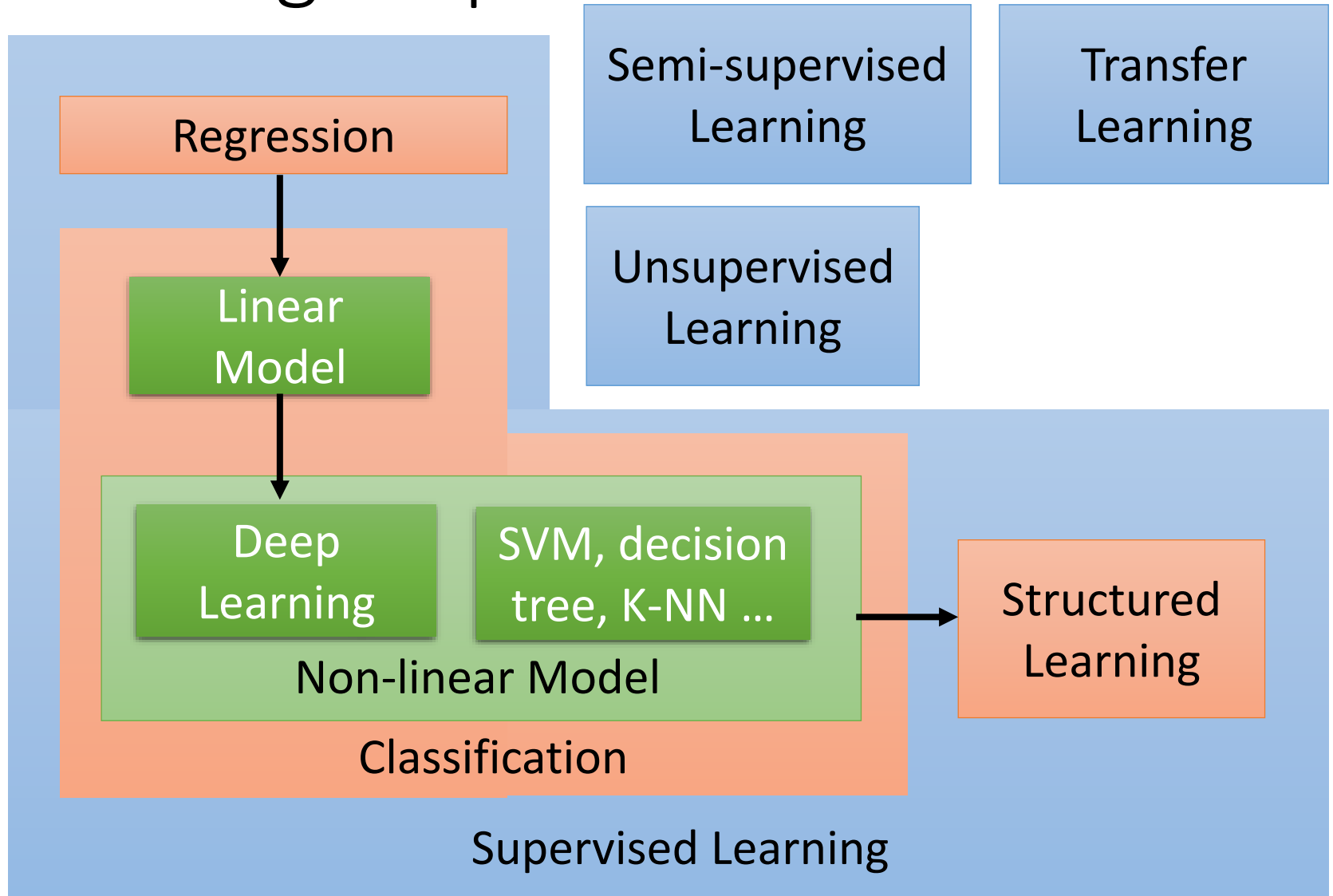


Haruhi



Data not related to the task considered
(can be either labeled or unlabeled)

Learning Map





Unsupervised Learning



Supervised

- Using historic data that has no labels.
- Discovers the intrinsic links of data.

Unsupervised

- Group photos into 20 groups based on their metadata.
- Segment customer profiles based on their demographics and purchase behavior.
- Find an anomaly in credit card usage patterns.

Semi-Supervised



Reinforcement

Unsupervised Learning



Learning Map

Unsupervised Learning

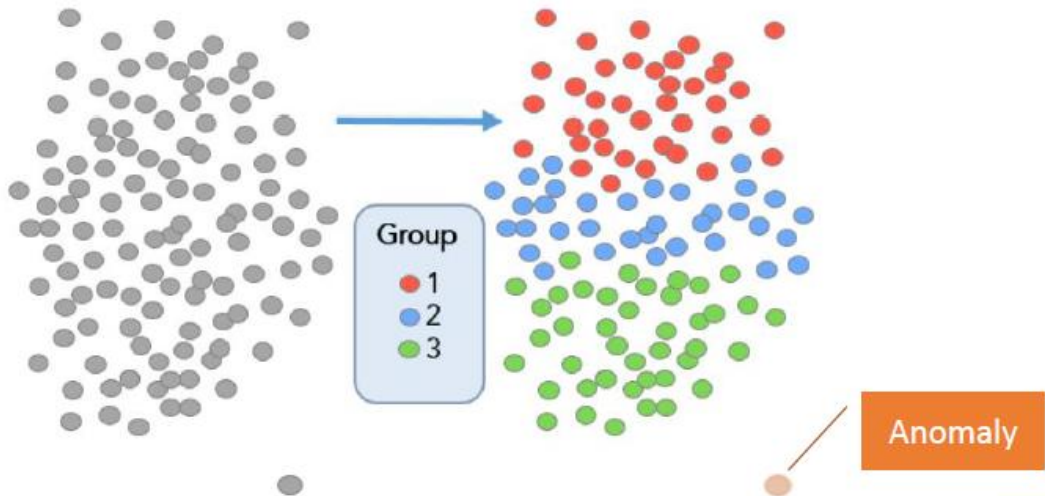
Supervised

Unsupervised

Semi-Supervised

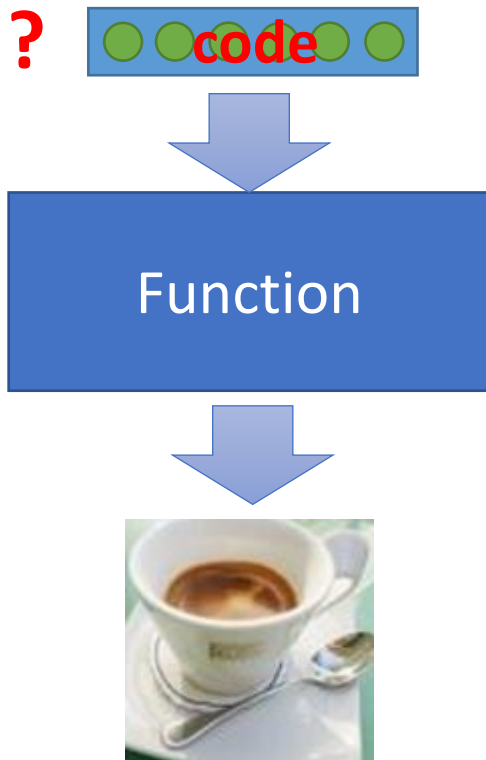
Reinforcement

- Useful for learning structure in the data (**clustering**) or detecting outliers (**anomaly**).



Unsupervised Learning

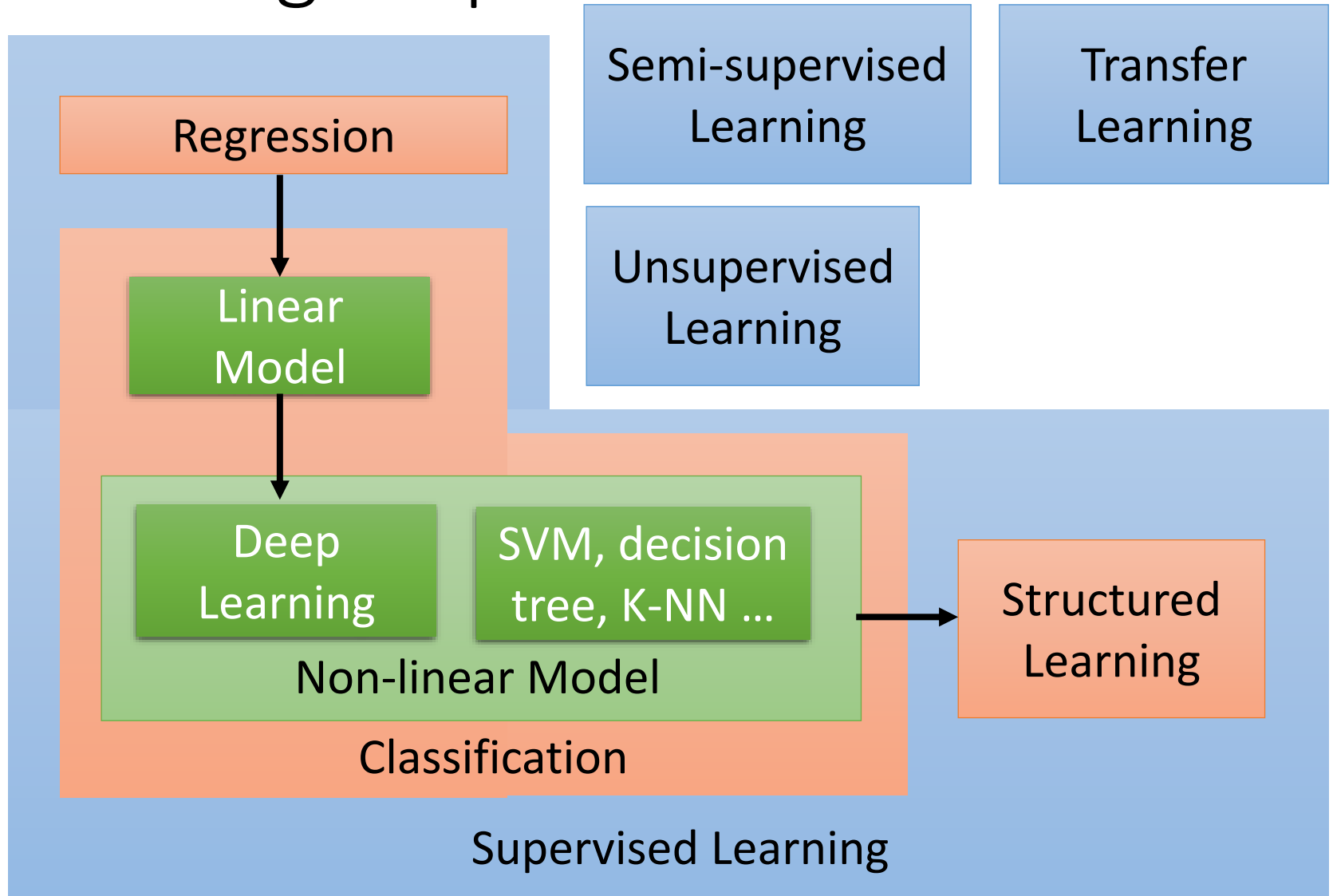
- Machine Drawing



Training data is a lot of images

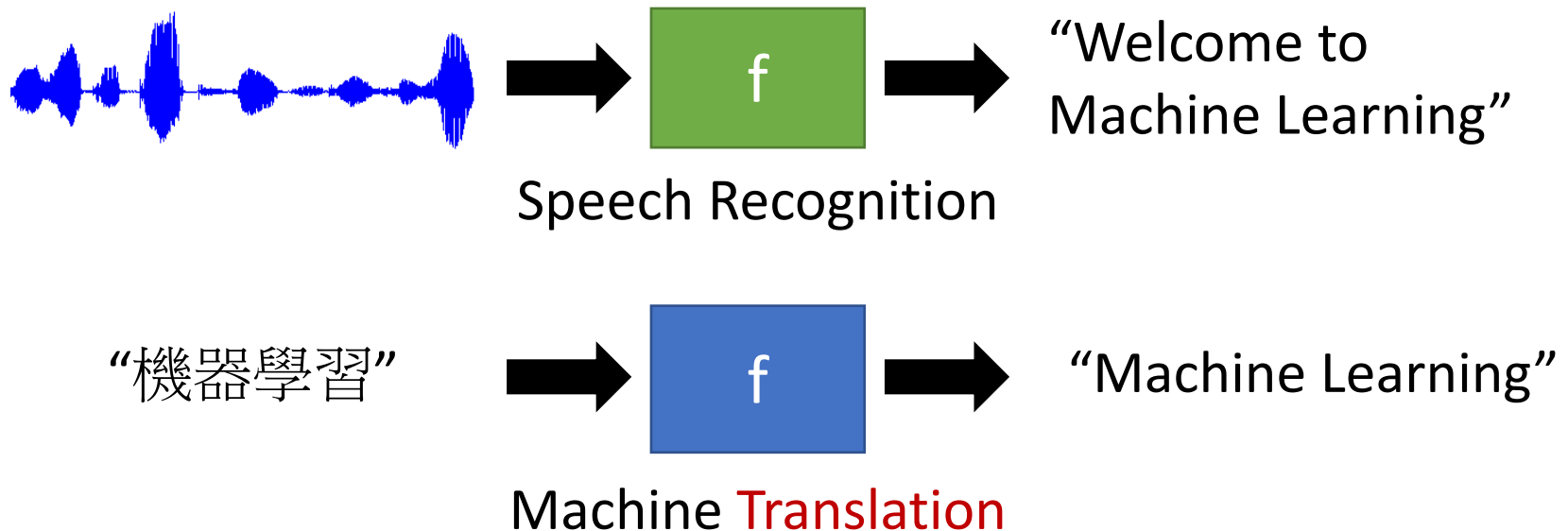


Learning Map



Structured Learning

- Beyond Classification



Face Recognition

Structured Learning

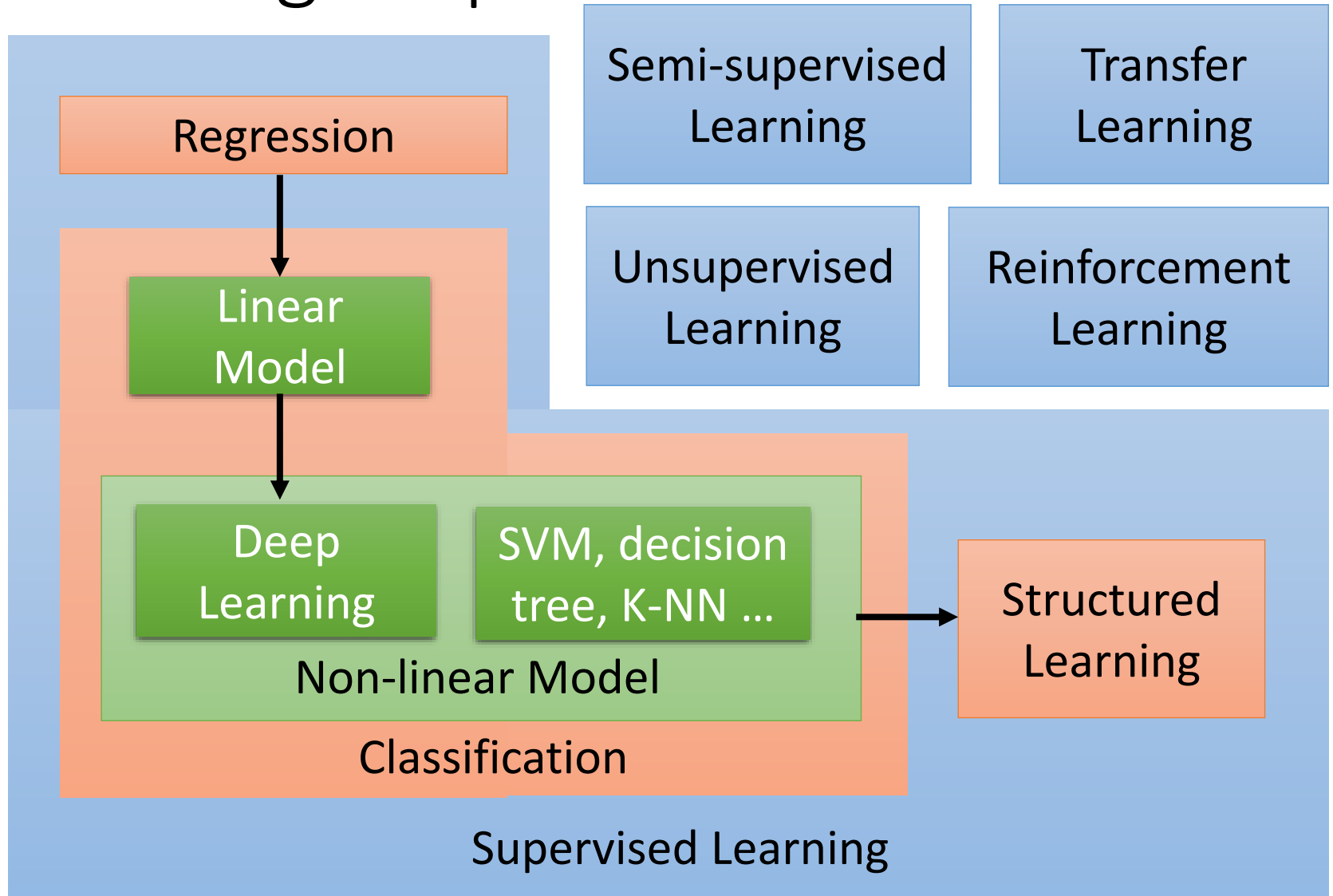
- Beyond Classification

Face Recognition

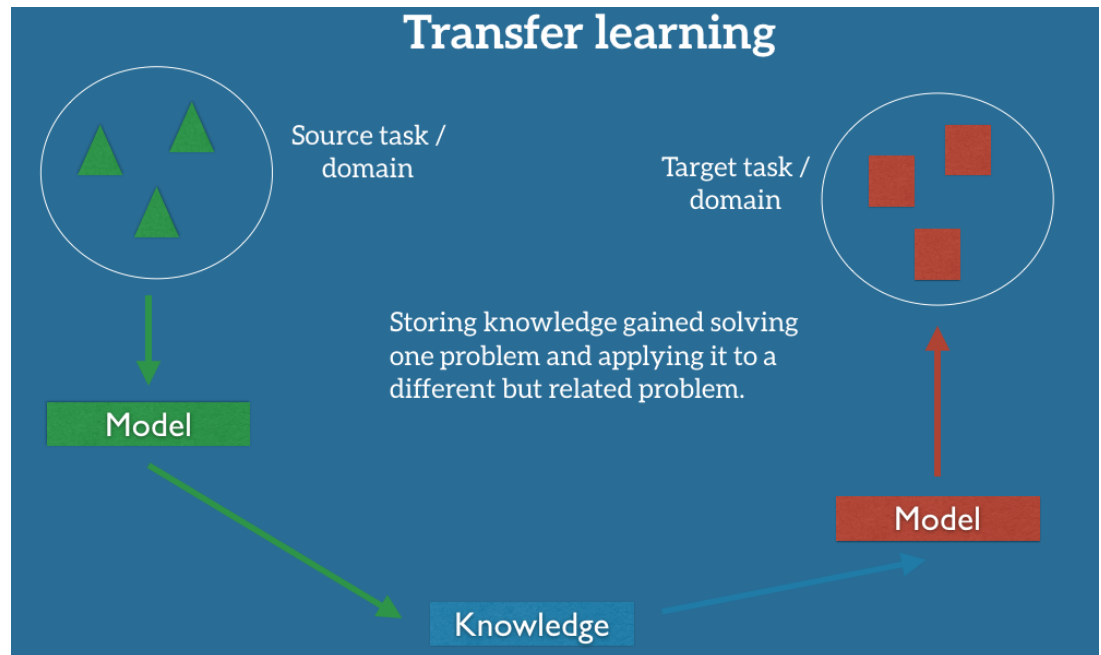
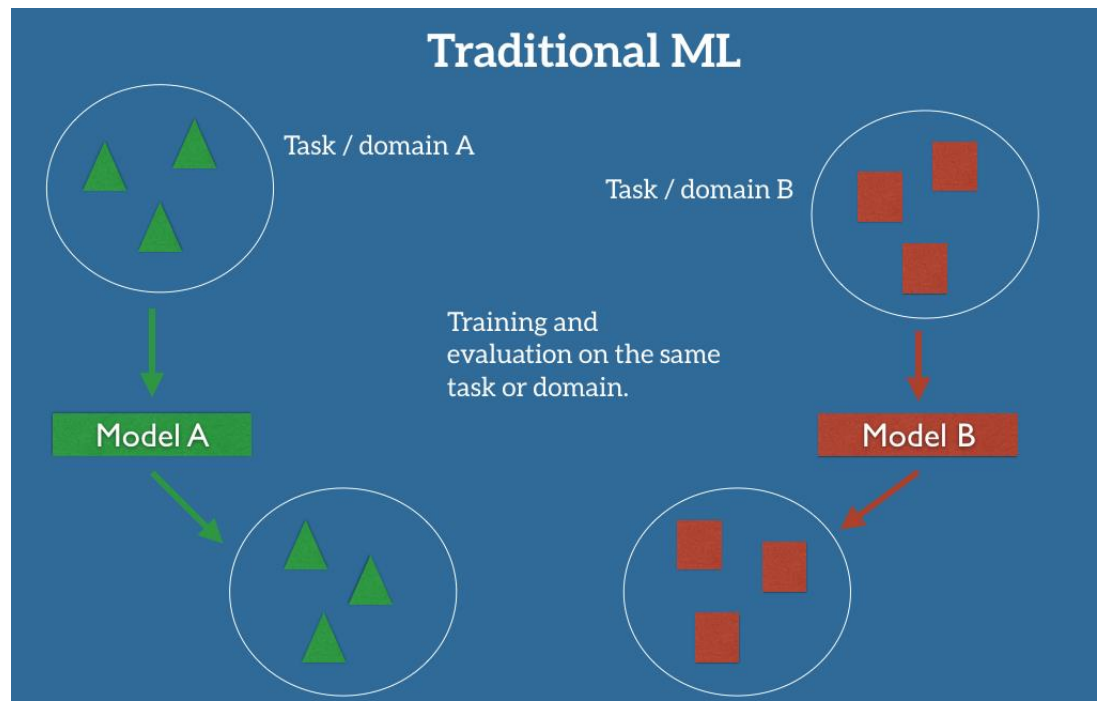


<https://github.com/yu4u/age-gender-estimation>

Learning Map

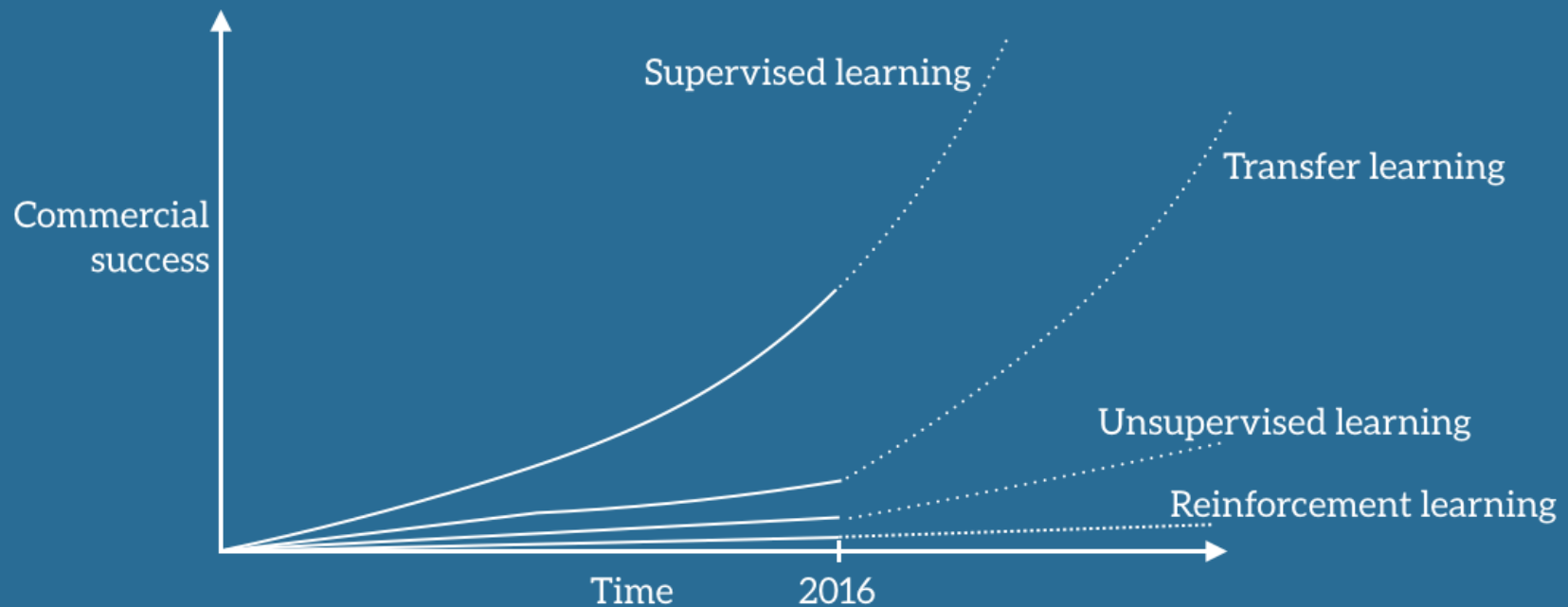


Transfer Learning - Machine Learning's Next Frontier



Transfer Learning - Machine Learning's Next Frontier

Drivers of ML success in industry



- Andrew Ng, NIPS 2016 tutorial

Transfer Learning - Machine Learning's Next Frontier



■ "Pure" Reinforcement Learning (cherry)

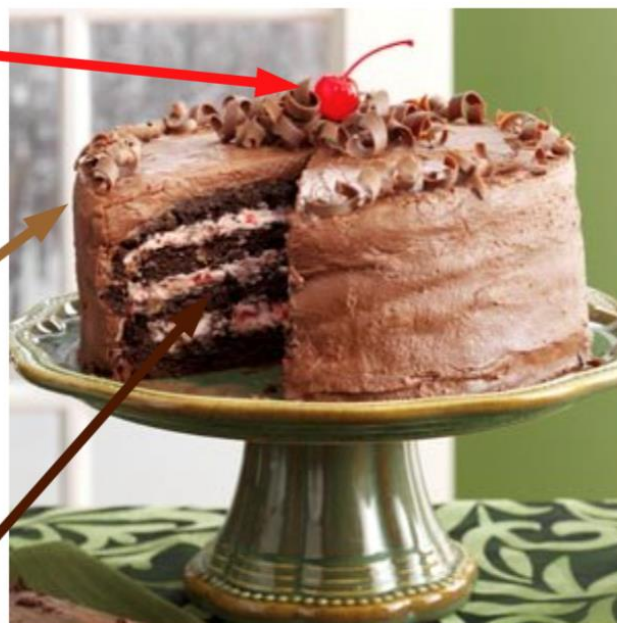
- ▶ The machine predicts a scalar reward given once in a while.
- ▶ **A few bits for some samples**

■ Supervised Learning (icing)

- ▶ The machine predicts a category or a few numbers for each input
- ▶ Predicting human-supplied data
- ▶ **10→10,000 bits per sample**

■ Unsupervised/Predictive Learning (cake)

- ▶ The machine predicts any part of its input for any observed part.
- ▶ Predicts future frames in videos
- ▶ **Millions of bits per sample**



■ (Yes, I know, this picture is slightly offensive to RL folks. But I'll make it up)

Reinforcement Learning





Reinforcement Learning



Supervised

Unsupervised

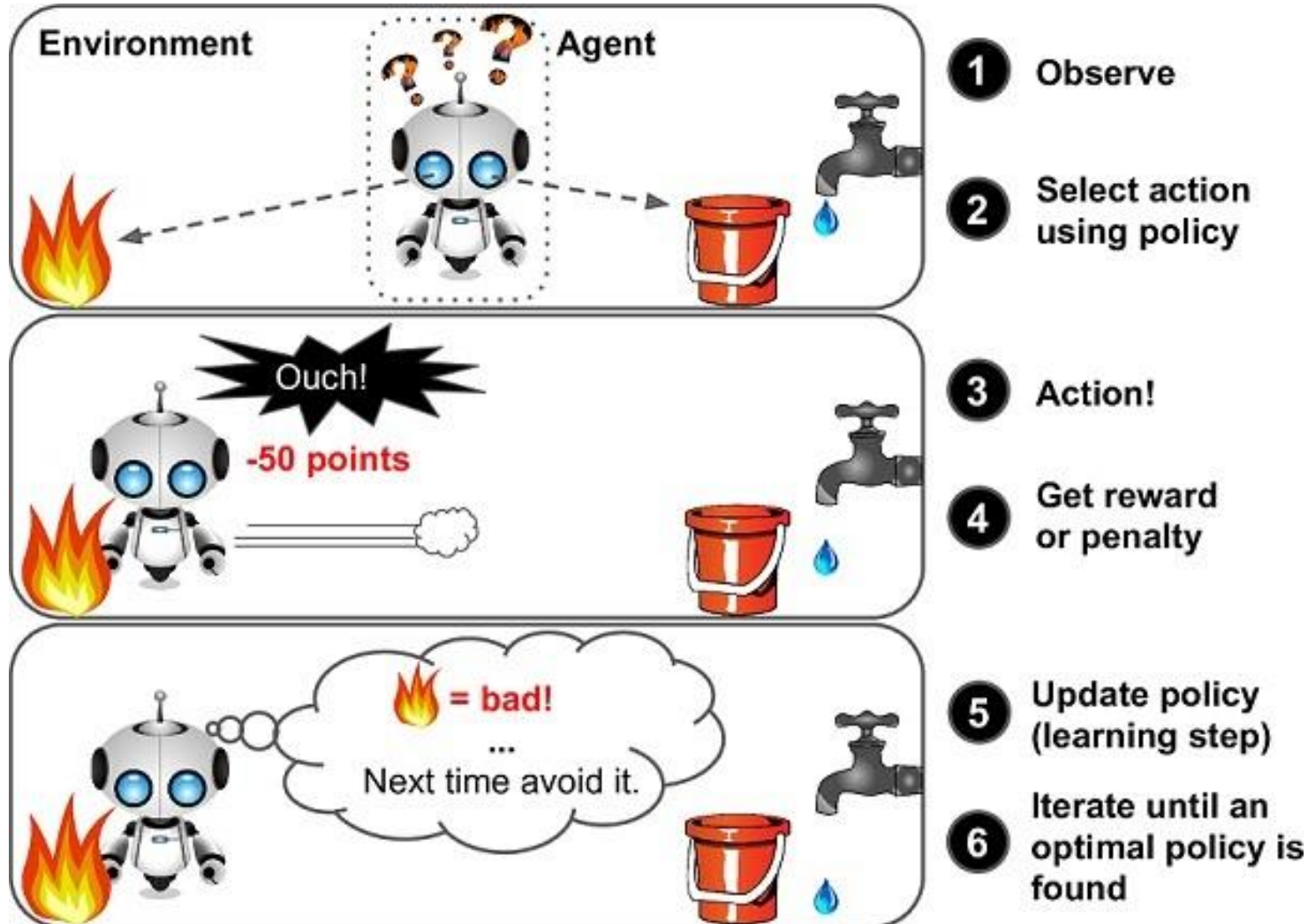
Semi-Supervised



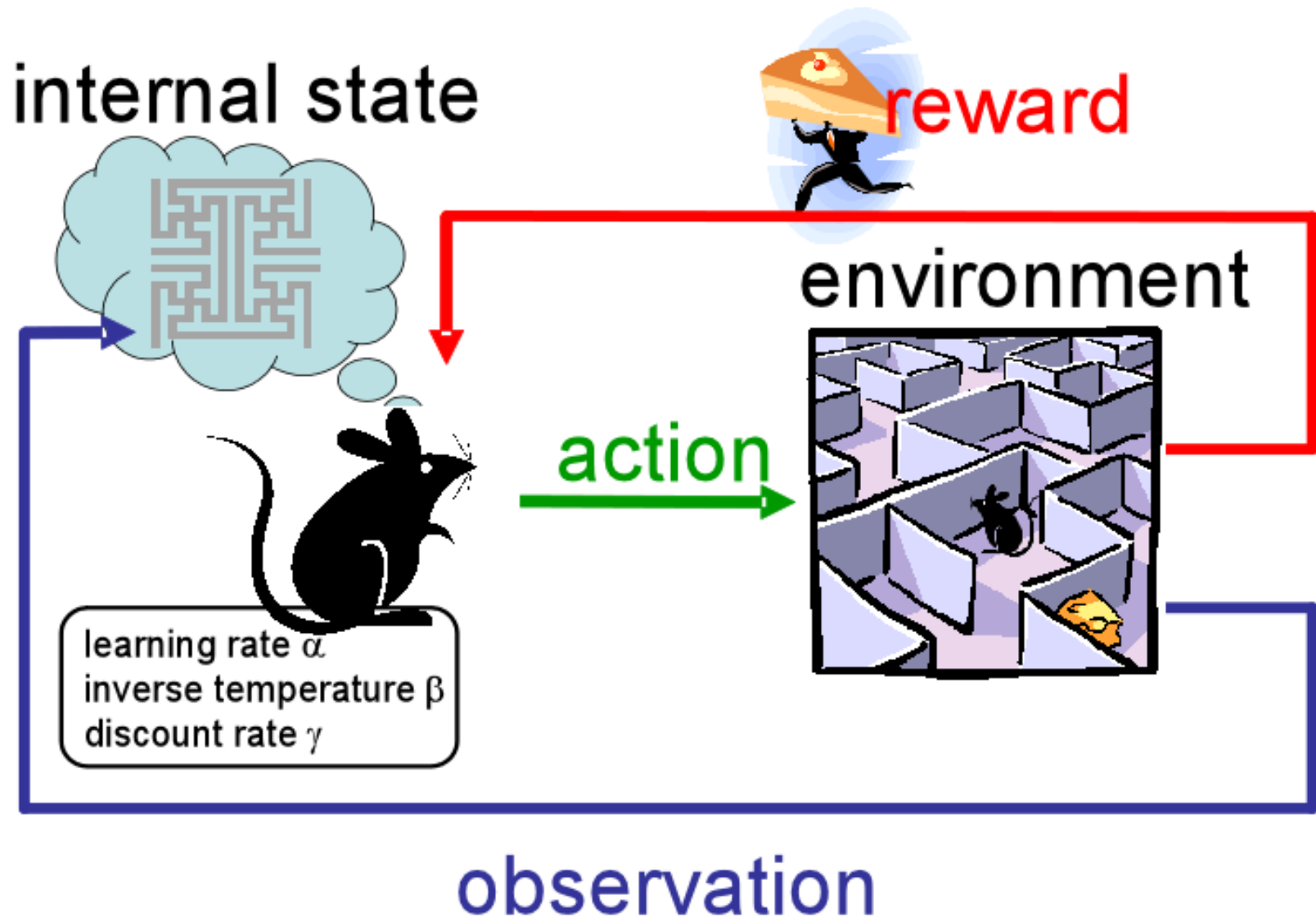
Reinforcement

- An agent interacts with an environment and watches the result of the interaction.
- Environment gives feedback via a positive or negative reward signal.
- The agent learns to optimize its interactions to maximize the reward.
 - An autonomous vehicle learns to put safety first, minimize ride time, and obey the rules of law.
 - An stock trading agent can decide to buy, sell or hold based on market status and transactions profit/loss.

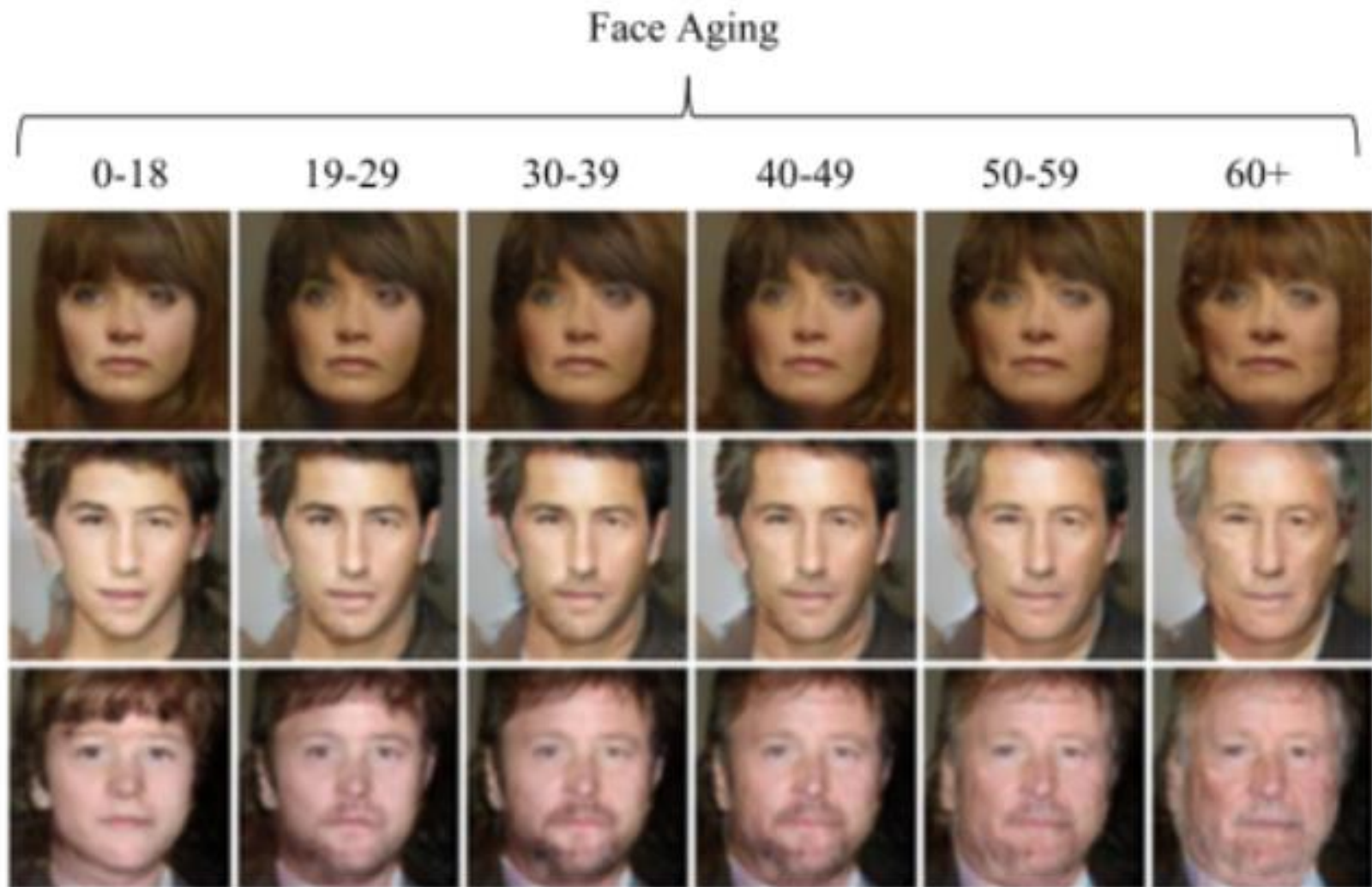
Reinforcement learning



Reinforcement learning



Face Aging



<https://medium.com/syncedreview/face-aging-with-conditional-generative-adversarial-networks-d41076379047>

Face Modeling: Generative Adversarial Neural Networks



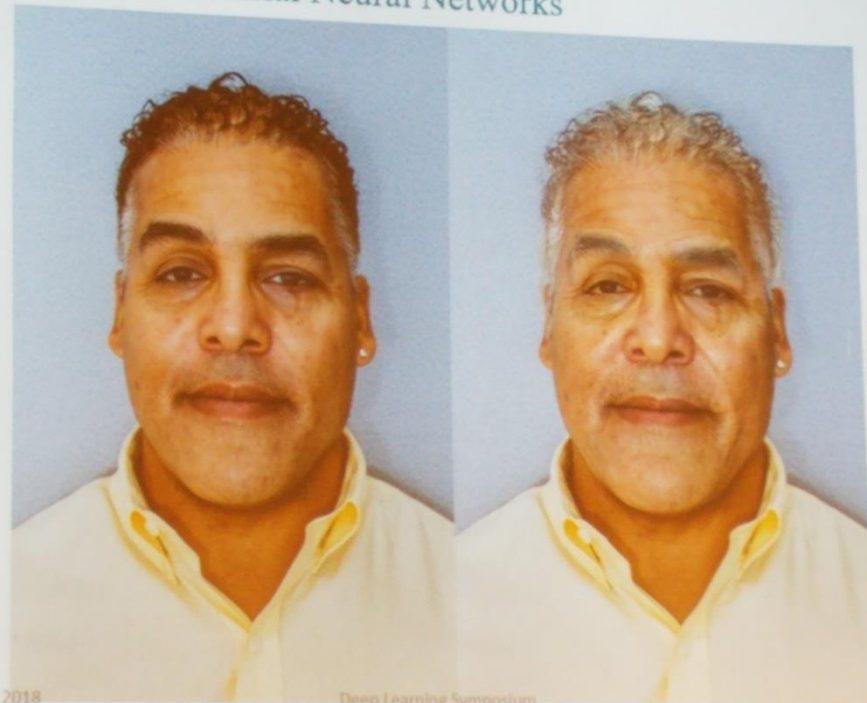
Sept 28 2018

Deep Learning Symposium

Face Modeling: Generative Adversarial Neural Networks



Face Modeling: Age Generative Adversarial Neural Networks



Sept 28 2018

Deep Learning Symposium

17

Face Aging

Supervised v.s. Reinforcement

- Supervised

Learning from
teacher



"Hello"

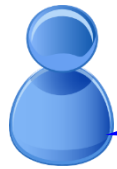
Say "Hi"



"Bye bye"

Say "Good bye"

- Reinforcement



.....



.....

.....

Hello 😊

Agent

.....

Agent

Learning from
critics



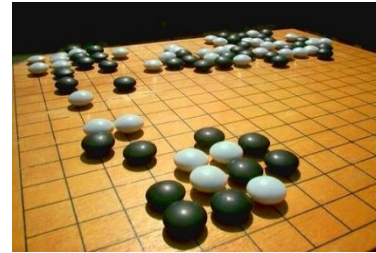
Bad

Supervised v.s. Reinforcement

- Supervised:



Next move:
"5-5"



Next move:
"3-3"

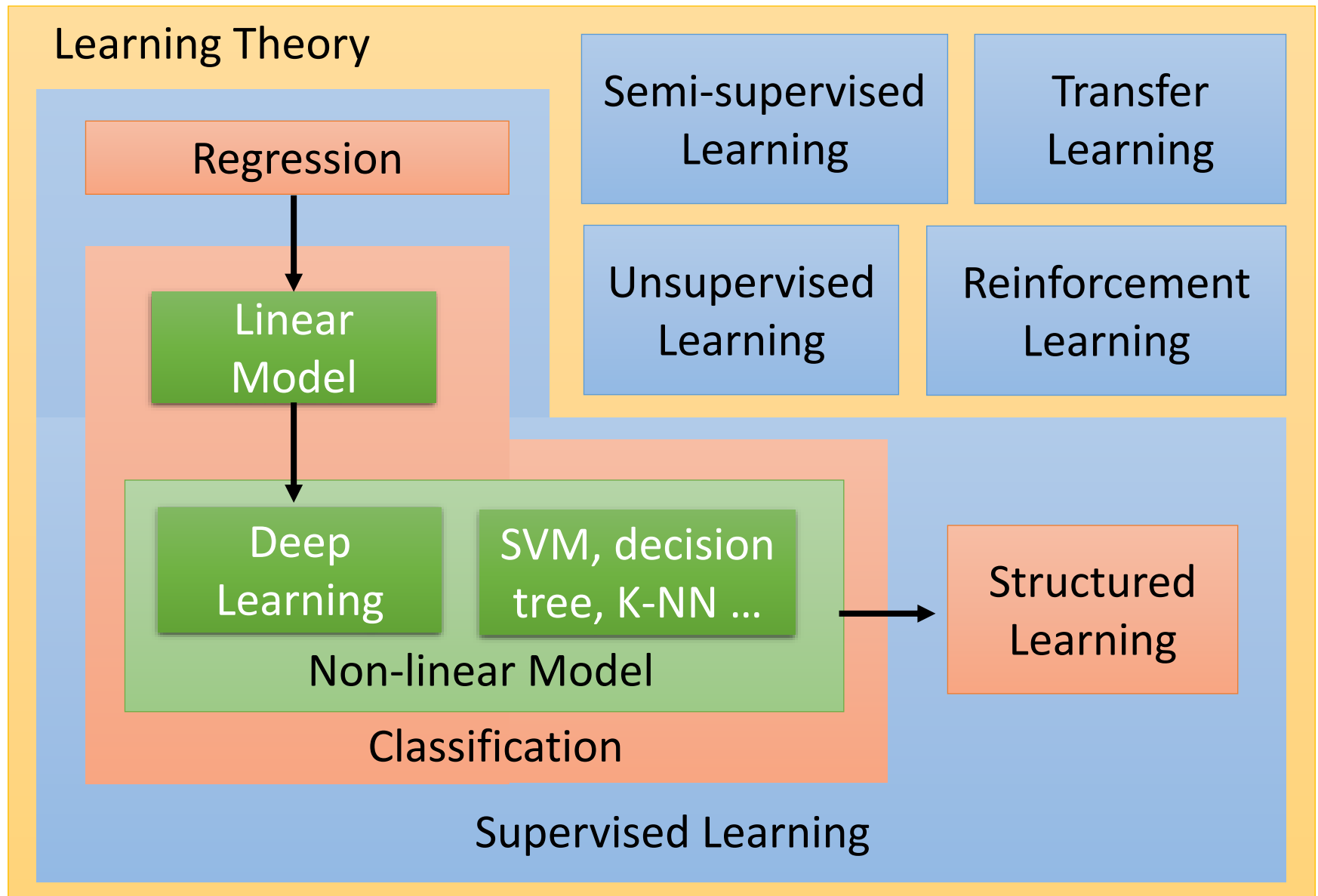
- Reinforcement Learning

First move → many moves → Win!

Alpha Go is supervised learning + reinforcement learning.

Learning Map

scenario task method





<http://www.express.co.uk/news/science/651202/First-step-towards-The-Terminator-becoming-reality-AI-beats-champ-of-world-s-oldest-game>

Why we need to learn Machine Learning?

Will AI replace our jobs? New job : AI trainer